Quantification: I534_z19c1-2
==========================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	20
interior lacuna count	20
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 20)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2815	2632	1594.62	px^2
major_axis_length_px	lacuna shape	86.4139	84.848	27.3025	px
minor_axis_length_px	lacuna shape	46.6552	44.321	13.8424	px
aspect_ratio	lacuna shape	1.9226	2.0925	0.5597	unitless
eccentricity	lacuna shape	0.7828	0.8784	0.1736	unitless
solidity	lacuna shape	0.745	0.7525	0.0994	unitless
roots_count	headline	11.75	11.5	4.3271	count
ring_length_r30_px	headline	309.4	325	105.222	px
ring_length_r60_px	reported	599.95	651	217.809	px
owned_length_px	ownership-dependent	864.355	752.062	648.616	px
edge_count	ownership-dependent	40.35	38.5	28.8084	count
mean_edge_length_px	ownership-dependent	20.8247	21.6932	3.0857	px
perimeter_px	normalised	288.247	279.099	98.758	px
ring_area_r30_px2	normalised	9678.3	9327	2520.39	px^2
ring_area_r60_px2	normalised	20407.5	21366	4636.47	px^2
in_frame_fraction_r30	normalised	0.9977	1	0.0105	unitless
in_frame_fraction_r60	normalised	0.9865	1	0.0417	unitless
ring_density_r30	normalised	0.0323747	0.0306134	0.0103555	px^-1
ring_density_r60	normalised	0.0292546	0.0278501	0.00894236	px^-1
roots_per_100px_perimeter	normalised	4.3094	4.4902	1.568	per 100 px
ring_attached_length_r30_px	network v2	281.65	282.5	98.1874	px
ring_attached_length_r60_px	network v2	467.65	514.5	196.029	px
ring_length_w_r30_px	network v2	352.61	377.055	119.37	px
ring_length_w_r60_px	network v2	682.781	743.683	247.629	px
sholl_crossings_r10	network v2	7	7	2.4709	count
sholl_crossings_r20	network v2	8.25	8.5	3.9719	count
sholl_crossings_r30	network v2	7.95	8.5	3.4409	count
loop_count	loops, report only	1.75	0.5	3.024	count
loop_edge_share	loops, report only	0.0967	0.0428	0.1244	share
ring_width_mean_r30_px	width	7.1946	7.1823	0.5605	px

Per field
----

measure	value	units
analysed_area_px2	845861	px^2
total_skeleton_length_px	22113	px
canalicular_length_density_per_px	0.0261426	px^-1
junction_count	433	count
junction_density_per_px2	0.0005119	px^-2
lacuna_count	20	count

lacunae_per_px2	2.364e-05	px^-2
skeleton_component_count	294	count
mean_component_length_px	75.2143	px
median_component_length_px	26.5	px
field_length_density_w_per_px	0.0296892	px^-1
field_density_without_flagged_per_px	0.0221754	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	319	px
unattached_component_count	19	count
canalicular_length_density_attached_per_px	0.0261426	px^-1
field_length_density_with_unattached_per_px	0.0265197	px^-1
excluded_area_px2	148874	px^2
analysed_field_area_px2	899702	px^2
loop_count	43	count
loop_edge_share	0.1271	share
edge_count	1235	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.9863	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	11.6619	px
width_pixels_used	22612	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	1823	56.4889	49.8735	1.1326	0.4696	0.8271	-1.1579	125.003	757.529	False
2	1637	57.8693	48.4179	1.1952	0.5477	0.7276	-0.0784	295.51	896.101	False
3	1291	47.4215	38.1837	1.2419	0.593	0.8741	0.4227	324.111	336.189	False
4	6026	129.945	61.3358	2.1186	0.8816	0.8485	-1.1185	377.349	511.866	False
5	1342	66.6735	36.4005	1.8317	0.8378	0.6286	1.399	396.13	787.283	False
6	2978	75.2858	55.7926	1.3494	0.6714	0.7831	-0.8753	453.94	606.7	False
7	4876	84.2608	78.5547	1.0726	0.3617	0.7525	0.8631	503.31	62.7231	False
8	3737	84.848	59.7745	1.4195	0.7097	0.8338	-0.4377	493.364	441.851	False
9	820	51.4532	24.5895	2.0925	0.8784	0.7614	1.0688	505.329	211.1	False
10	3553	104.824	44.321	2.3651	0.9062	0.8721	0.2384	581.822	856.943	False
11	4782	122.937	62.4014	1.9701	0.8616	0.6511	1.4416	592.132	630.596	False
12	1615	90.9802	31.7451	2.866	0.9372	0.5886	-0.9716	606.27	724.946	False
13	2632	101.535	38.4077	2.6436	0.9257	0.6745	-1.3449	587.391	245.703	False
14	5166	140.866	61.0892	2.3059	0.9011	0.6428	0.013	686.162	69.5337	False
15	1712	87.747	36.8261	2.3827	0.9077	0.6786	-0.5403	724.265	671.19	False
16	914	54.9878	33.7201	1.6307	0.7899	0.5745	-0.9811	788.83	430.015	False
17	2692	86.6013	40.7838	2.1234	0.8822	0.8628	-1.1084	820.32	99.4658	False
18	1680	84.8432	32.3344	2.6239	0.9245	0.7391	-0.8417	827.977	860.805	False
19	4209	112.297	51.8973	2.1638	0.8868	0.8336	-0.6033	855.455	275.629	False
20								402.815	659.098	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	5	198	622	864.051	38	22.7382	6.2345
2	6	157	316	436.061	20	21.803	6.4455
3	10	339	797	2697.91	125	21.5833	6.341
4	21	398	668	707.884	43	16.4624	7.0146
5	12	462	989	1551.4	75	20.6853	7.4784
6	9	193	228	254.522	13	19.5786	7.0148
7	17	461	696	922.536	40	23.0634	7.2005
8	14	257	538	348.978	19	18.3672	7.2011
9	13	385	605	796.24	49	16.2498	8.3586
10	17	384	985	2167.2	92	23.5565	7.6189
11	15	366	587	701.411	35	20.0403	7.6586
12	7	172	258	116.841	9	12.9823	7.9461
13	15	392	654	871.08	39	22.3354	7.1641
14	16	436	680	578.801	26	22.2616	7.6003
15	10	357	648	1127.39	49	23.0079	7.1053
16	12	252	655	1106.84	43	25.7406	7.1218
17	11	274	433	381.333	20	19.0667	6.7084
18	9	284	673	582.517	24	24.2715	7.8693
19	11	311	742	915.649	40	22.8912	6.5456
20	5	110	225	158.468	8	19.8085	7.2647

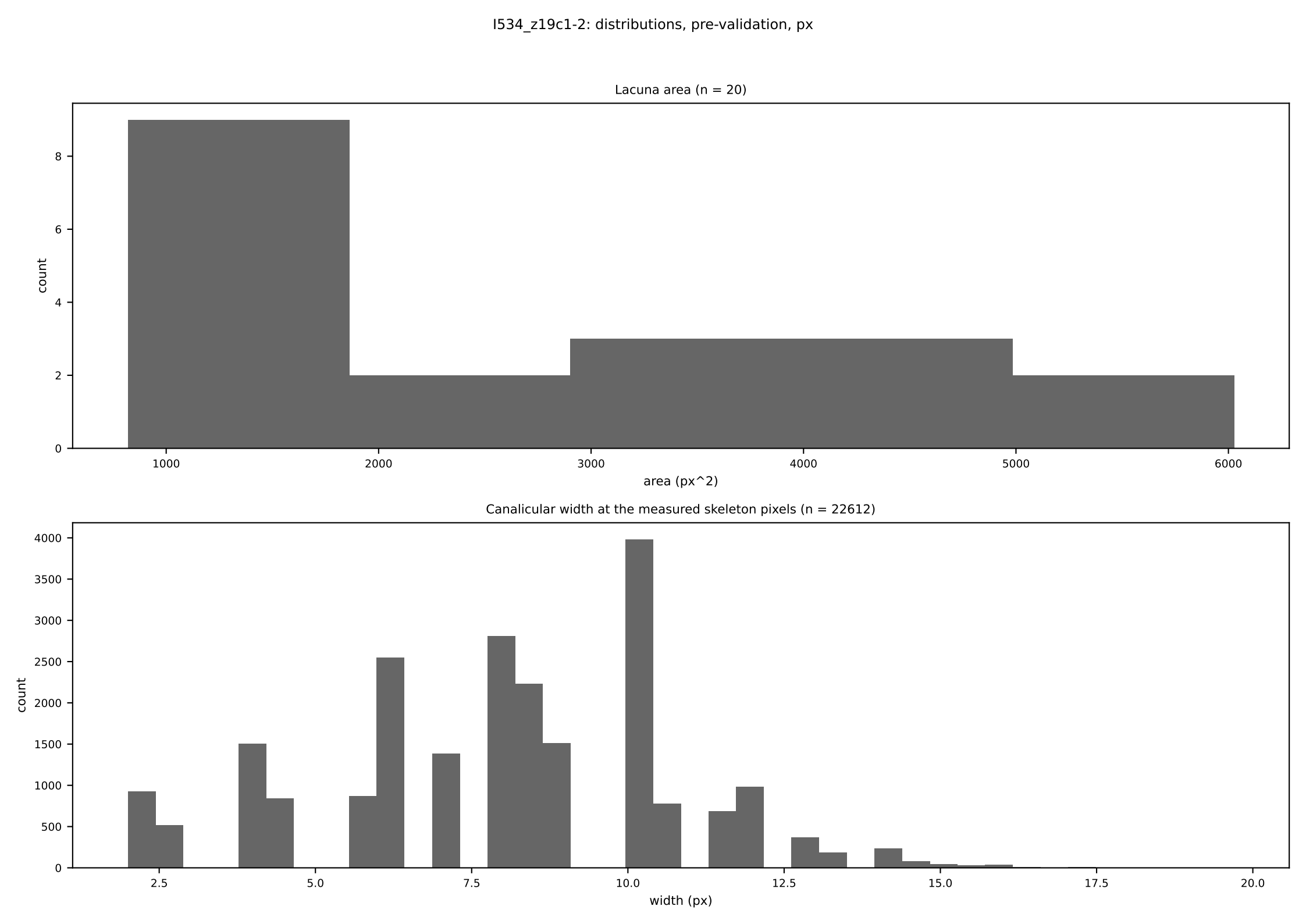
Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	272.936	8396	22205	1	1	0.0235827	0.0280117	1.8319
2	283.463	8668	21456	1	1	0.0181126	0.0147278	2.1167
3	164.51	7220	19561	1	1	0.0469529	0.0407443	6.0787
4	382.854	12834	23401	1	1	0.0310114	0.0285458	5.4851
5	262.871	8654	20100	1	1	0.0533857	0.049204	4.565
6	274.735	8747	13031	1	1	0.0220647	0.0174967	3.2759
7	375.103	12127	21219	0.9532	0.8444	0.0380144	0.0328008	4.5321
8	300.049	10385	21896	1	1	0.0247472	0.0245707	4.6659
9	155.717	6688	13329	1	1	0.0575658	0.0453897	8.3485
10	288.735	10625	24527	1	1	0.0361412	0.0401598	5.8878
11	439.83	12113	21276	1	1	0.0302155	0.0275898	3.4104
12	289.848	8213	14035	1	1	0.0209424	0.0183826	2.4151
13	316.777	10912	23620	1	1	0.0359237	0.0276884	4.7352
14	533.263	15650	25399	1	0.8882	0.0278594	0.0267727	3.0004
15	263.664	9930	21920	1	1	0.0359517	0.029562	3.7927
16	215.723	8071	20618	1	1	0.0312229	0.0317684	5.5627
17	247.279	9530	20421	1	0.9977	0.0287513	0.0212037	4.4484
18	251.22	9124	23689	1	1	0.0311267	0.0284098	3.5825
19	352.617	11633	27463	1	1	0.0267343	0.0270182	3.1195
20	93.7401	4046	8984	1	1	0.0271873	0.0250445	5.3339

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	183	429	233.108	722.796	6	9	10
2	128	177	176.752	350.392	4	2	5
3	285	555	385.363	910.95	6	14	10
4	357	560	446.676	752.541	9	12	10
5	453	828	526.375	1124.06	11	13	12
6	168	191	225.208	263.593	3	4	1

7	428	634	528.831	787.512	11	12	10
8	228	319	292.421	609.6	6	7	5
9	341	541	434.635	679.487	8	12	7
10	373	829	429.877	1112.34	8	10	14
11	339	424	410.191	668.127	8	6	6
12	172	246	195.167	297.735	4	3	4
13	352	543	447.889	746.24	7	11	8
14	353	499	498.174	778.039	10	13	14
15	326	557	415.161	743.34	9	10	9
16	251	607	291.007	744.027	6	4	6
17	266	354	309.078	489.76	8	8	7
18	260	530	312.966	746.085	7	7	9
19	280	426	368.747	869.891	7	7	9
20	90	104	124.569	259.108	2	1	3


I534_z19c1-2: distributions, pre-validation, px
Lacuna area (n = 20)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 22612)
count
width (px)